Abu Dhabi City Campus Daily/Weekly Planner

Week 10: 2nd – 6th November 2026

Subject: Mathematics **Unit Title: Differentiation** **Grade: 12**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1: Average Rate & Instantaneous Rate of Change	I can calculate and interpret average and instantaneous rates of change using graphs, tables, equations, and real-world contexts.	2.1 Clarify mathematical vocabulary and notation using graphs, tables, worked examples, and connections. 2.3 Support decoding of rate-of-change notation using multiple representations.	Related Videos / Resources Calculus AB/BC – 2.1 Defining Average and Instantaneous Rate of Change at a Point Calculus AB/BC – 2.2 Defining the Derivative of a Function and Using Derivative Notation Calculus AB/BC – 2.5 Applying the Power Rule
Lesson 2: Defining the Derivative	I can explain the derivative as the limit of average rates of change and use the limit definition to find the derivative at a point.	2.3 Support decoding of the difference quotient through annotated worked examples. 5.3 Build fluency with graduated scaffolds and guided practice.	
Lesson 3: Definition of the Derivative – Practice & Interpretation	I can apply the limit definition to calculate f'(a) and interpret the result as the slope of the tangent line and instantaneous rate of change.	5.3 Build fluency through worked examples and independent practice. 8.2 Extend learning with unfamiliar polynomial functions and explanation tasks.	
Lesson 4: Power Rule of Differentiation	I can use the power rule to differentiate power functions and simplify derivative expressions accurately.	8.2 Vary challenge level through differentiated derivative problems. 9.3 Check answers by comparing symbolic results with graphs or GDC estimates.	
Homework	Complete assigned practice on average/instantaneous rates, the limit definition of the derivative, and the power rule. Resources posted on Teams/OneNote.		
Teacher's notes/ Reminders	Students should bring a GDC/graphing calculator, notebook and device. Check understanding of limit notation before progressing to derivative calculations.		

Technology Integration	Microsoft Teams; OneNote; GDC/graphing calculator; Desmos; lesson PowerPoints; digital practice and assessment materials.
Cross-curricular links	**Physics/Science:** interpret rates of change in motion, velocity and changing quantities. **Economics/real-world modelling:** interpret marginal or instantaneous change.
External Assessment Integration	Direct alignment with AP Calculus AB

Wellbeing/ ATL Skills	**Thinking Skills:** Connect numerical, graphical and analytical representations; select appropriate methods and analyse errors. **Communication Skills:** Use correct limit and derivative notation and explain reasoning clearly. **Self-Management Skills:** Organise multi-step algebra, practise independently, check work and reflect on misconceptions.
Standards	AP Calculus AB: 2.1 Defining Average and Instantaneous Rates of Change at a Point; 2.2 Defining the Derivative of a Function and Using Derivative Notation; 2.5 The Power Rule.
Inquiry/ TOK Question	How does a mathematical limit allow us to define instantaneous change, and when should we trust a mathematical model to represent change in the real world?
Success Criteria (Measurable)	**Lesson 1: Average Rate & Instantaneous Rate of Change** • **All students must** calculate average rate of change over an interval and identify instantaneous rate of change at a point from a graph, table or function. • **Most students should** connect instantaneous rate of change to the slope of a tangent and explain how it differs from average rate of change. • **Some students could** explain how secant slopes approach the tangent slope as the interval shrinks. **Lesson 2: Defining the Derivative** • **All students must** identify the difference quotient and use correct derivative notation. • **Most students should** use the limit definition to calculate f'(a) for polynomial functions, showing algebraic steps. • **Some students could** use the limit definition to derive a general expression for f'(x) and explain its graphical meaning. **Lesson 3: Definition of the Derivative – Practice & Interpretation** • **All students must** substitute into the difference quotient and simplify a basic limit-definition problem. • **Most students should** evaluate the limit to find f'(a) and interpret the result as a tangent slope or instantaneous rate. • **Some students could** compare different methods/representations and identify and correct errors in a worked solution. **Lesson 4: Power Rule of Differentiation** • **All students must** apply the power rule to differentiate a single power function, including constant multiples. • **Most students should** differentiate sums and differences of polynomial terms and simplify the result accurately. • **Some students could** solve unfamiliar polynomial differentiation problems and justify the use of the power rule.
National Identity Domain	Respect الاحترام Area: Citizenship Implementation: Promote mathematical responsibility by showing complete working, using precise notation, and checking the reasonableness of derivative results.
	Area: Citizenship
	Implementation: Encourage persistence, honest self-assessment and evidence-based explanations when students encounter errors in multi-step calculations.